

Homework 3

Problem 1. For natural numbers $m \leq n$ calculate (i.e. express by a simple formula not containing a sum) $\sum_{k=m}^n \binom{k}{m} \binom{n}{k}$.

Solution. $\binom{k}{m} \binom{n}{k} = \frac{k!}{m!(k-m)!} \cdot \frac{n!}{k!(n-k)!} = \frac{n!}{m!(n-m)!} \cdot \frac{(n-m)!}{(k-m)!(n-k)!} = \binom{n}{m} \binom{n-m}{n-k}$.

Thus $\sum_{k=m}^n \binom{k}{m} \binom{n}{k} = \sum_{k=m}^n \binom{n}{m} \binom{n-m}{n-k} = \binom{n}{m} \sum_{k=m}^n \binom{n-m}{n-k} = \binom{n}{m} 2^{n-m}$. □

Problem 2. Count the number of linear extensions for the following partial ordering:

X is a disjoint union of sets $X_1, X_2, \dots, X_k$ of sizes $r_1, r_2, \dots, r_k$, respectively. Each X_i is linearly ordered by $\leq$, and no two elements from the different X are comparable.

Solution. $\binom{r_1+r_2+\dots+r_k}{r_1, r_2, \dots, r_k}$. □

Problem 3. Given a set X with $|X| = n$, determine the number of ordered set pairs $\langle A, B \rangle$ where $A \subseteq B \subseteq X$.

Solution. As a set B of size b has exactly 2^b subsets A , the number of such ordered pairs is

$$\sum_{0 \leq b \leq n} 2^b \binom{n}{b} = (1+2)^n = 3^n.$$

□

Problem 4. Determine the coefficient of x^3 in $(2+x)^{3/2}/(1-x)$

Solution.

$$\begin{aligned} &= (x+2)^{3/2}(1+x+x^2+\dots) \\ &= \sum_{k=0}^{\infty} \binom{3/2}{k} x^k (2)^{3/2-k} (1+x+x^2+\dots) \end{aligned}$$

The coefficient of x^3 is $\sum_{k=0}^3 \binom{3/2}{k} (2)^{3/2-k}$. Then use the Newton formula

□